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## **SUBJ: HIF AFB Standard Operating Procedures**

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This document establishes the Hill Air Force Base Non-Federal Contract Air Traffic Control Tower (HIF NFCT) Standard Operating Procedures within the Salt Lake City ARTCC on VATSIM (vZLC). This SOP establishes standardized air traffic control, flight operations, and airfield management procedures at Hill Air Force Base (HAFB). It ensures safe, efficient, and coordinated air traffic operations by detailing responsibilities, communication protocols, and operational guidelines. The provisions and procedures described herein are supplemental to vZLC Facility Policies and FAA Order JO 7110.65.

The information contained herein will be used only for flight simulation purposes on the VATSIM network. It is not intended, nor should it be used for real-world navigation. The Virtual Salt Lake City ARTCC is not affiliated with the FAA, DOD, Salt Lake City ARTCC, or any governing aviation body.

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## Record of Changes

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# **Chapter 1. Standard Operating Control Procedures**

## **Section 1. Introduction**

### **1-1-1. Purpose**

This document establishes the Hill Air Force Base Non-Federal Contract Air Traffic Control Tower (HIF NFCT) Standard Operating Procedures within the Salt Lake City ARTCC on VATSIM (vZLC). This SOP establishes standardized air traffic control, flight operations, and airfield management procedures at Hill Air Force Base (HIF). It ensures safe, efficient, and coordinated air traffic operations by detailing responsibilities, communication protocols, and operational guidelines.

### **1-1-2. Audience**

All vZLC controllers and visitors contained within the vZLC and VATUSA roster.

### **1-1-3. Distribution**

This document is authorized for unrestricted use and release and is available in the Resources Section of the vZLC Website.

### **1-1-4. Cancellation**

N/A

## Section 2. Operational Positions

Table 1-2.1. HIF NFCT Operational Positions Table

| Sector/Position         | STARS ID | VATSIM Callsign | Frequency |
|-------------------------|----------|-----------------|-----------|
| Clearance Delivery (CD) | 4D       | HIF_DEL         | 124.100   |
| Ground Control (GC)     | 4G       | HIF_GND         | 121.600   |
| Local Control (LC)      | 4L       | HIF_TWR         | 127.150   |

**Bold** designates a primary position.

## Section 3. Intrafacility Coordination Procedures

### 1-3-1. Runway Configurations

The active runway configuration refers to the general cardinal direction of departures and arrivals currently in use. Unless otherwise coordinated with Salt Lake City TRACON (S56), the active runway must be selected in accordance with the following:

- (1) When the tailwind is component 10 knots or greater for Runway 14, Runway 32 should be used.
- (2) When the tailwind component exceeds 10 knots for Runway 32 or the weather is below circling minima, Runway 14 should be used.
- a. South Flow is designated as the “calm wind” runway configuration.
- b. South Flow: Landing and departing Runway 14.
- c. North Flow: Landing and departing Runway 32.

### 1-3-2. Changing Runway Configurations

The Local Control (LC) position is the final authority for selecting the runway configuration, but coordination with GC and TRACON must be accomplished prior to changing the runway configuration.

- a. LC must:
  - (1) Coordinate with GC and S56 to determine which aircraft will be the last departure and arrival in the old runway configuration.
  - (2) Ensure that:
    - (a) The Tower Display Workstation (STARS) display is adjusted to indicate the correct flow.
    - (b) The ATIS is updated to reflect the runway configuration change.
- b. S56 TRACON must:
  - (1) Ensure the STARS display is adjusted to indicate the correct flow.
  - (2) Notify S56 TRACON Hart And Bear Sectors (or other appropriate overlying sector) of the runway configuration change.

### 1-3-3. ATIS

Regardless of the position broadcasting the ATIS, the following statements must be included on the ATIS broadcast:

- a. During VFR weather conditions: “DEPARTING VFR AIRCRAFT ADVISE GROUND CONTROL OF REQUESTED HEADING OR DESTINATION.”
- b. When the outside air temperature is 26°C or greater: “CHECK DENSITY ALTITUDE.”

#### **NOTE-**

*This statement must be made immediately after stating the altimeter setting.*

## Section 4. Opposite Direction Operations (ODO)

### 1-4-1. General

Opposite Direction Operations consist of IFR/VFR operations conducted to the same or parallel runway where an aircraft is operating in a reciprocal direction of another aircraft arriving, departing, or conducting an approach.

#### **NOTE-**

*Departing aircraft include aircraft conducting a touch-and-go, stop-and-go, low approach, missed approach, or go-around.*

### 1-4-2. Responsibilities

- a. LC and Hart (S56) share the responsibility to coordinate ODO and issue traffic advisories (see below).
- b. LC is responsible for applying the cutoff point between opposite direction arriving and departing aircraft.
- c. Hart is responsible for applying the cutoff point between opposite direction arrival aircraft.

### 1-4-3. IFR Aircraft Procedures

- a. General:
  - (1) Traffic advisories must be issued to all ODO aircraft.

#### **PHRASEOLOGY-**

*OPPOSITE DIRECTION TRAFFIC (distance) MILE FINAL, (type aircraft).*

*OPPOSITE DIRECTION TRAFFIC DEPARTING RUNWAY (number), (type aircraft).*

*OPPOSITE DIRECTION TRAFFIC (position), (type aircraft).*

- (2) VFR aircraft conducting practice approaches must receive IFR services.
  - (3) Opposite direction, same or parallel runway, operations with opposing traffic inside the cutoff point are prohibited.
  - (4) Lateral separation must exist before any other type of separation is applied.
- b. Coordination:
  - (1) LC must verbally request opposite direction departures with S56.
  - (2) S56 must verbally request opposite direction arrivals with LC.
  - (3) Initial coordination must include callsign, type, and arrival/departure runway. The phrase "opposite direction" must be used.
- c. Cutoff procedures for aircraft conducting ODO to the same runway:

#### **NOTE-**

*When both aircraft are receiving IFR services, visual separation is not authorized.*



- (1) A departing aircraft must be airborne and have turned to avoid conflict prior to an aircraft reaching:
  - (a) A point five flying miles from the threshold of the runway of intended landing or, if an aircraft is established in the traffic pattern, prior to that aircraft turning base leg.
  - (b) If the above conditions are not met, action must be taken to ensure control instructions are issued to protect the integrity of the cutoff points.
- (2) An arriving aircraft must cross the runway threshold prior to an aircraft reaching:
  - (a) A point five flying miles from the threshold of the runway of intended landing or, if an aircraft is established in the traffic pattern, prior to that aircraft turning base leg.
  - (b) If the above conditions are not met, action must be taken to ensure control instructions are issued to protect the integrity of the cutoff points.
- d. Cutoff procedures for aircraft conducting ODO to parallel runways:
  - (1) Provide a turn away from the opposing arriving traffic before the arrival is inside of the cutoff point to the other runway. The same cutoff points in subparagraph c. must apply.

**NOTE-**

- 1. Visual separation of any type may be applied after the turn away from opposing traffic is issued.*
- 2. Prior to a pilot providing visual separation, a heading with at least 15° angular difference between the aircraft's flight paths must be assigned to ensure flight paths do not cross.*

**1-4-4. VFR Aircraft With ODO Operations In Effect Procedures**

- a. Ensure VFR aircraft are issued a turn to avoid conflict with the opposing IFR/VFR traffic.
- b. During coordination, the phrase "opposite direction" must be used.
- c. LC must issue traffic information to both aircraft and indicate the direction that the departure will turn (arrival/departure ODO) or the location of the opposing aircraft on final (arrival/arrival ODO).

## **Chapter 2: Tower Operational Positions**

### **Section 1. Clearance Delivery**

#### **2-1-1. Responsibilities**

- a. Monitor the CD frequency (124.100/335.80 MHz).
- b. Prepare and update flight progress strips in accordance with the ZLC Flight Strip Handling Directive (7210.3).
- c. Issue departure clearances to IFR aircraft in accordance with posted traffic management initiatives (TMIs), letters of agreement (LOAs), and published preferred routes.
- d. Issue departure clearances to VFR aircraft.
- e. Assume other clearance delivery or other cab responsibilities, as assigned.

#### **2-1-2. IFR Departure Routing**

IFR departure aircraft must be routed in accordance with the following procedures:

- a. Except when the flight crew indicates they are unable to accept, all aircraft must be assigned the following SIDs depending on the departure runway:
  - (1) Runway 14: FRMNT#. The DEVLN# may be issued if filed by the aircraft.
  - (2) Runway 32: WLLRD#. The DEVLN# may be issued if filed by the aircraft.
- b. If the flight crew is unable to accept a SID, the aircraft must be instructed to fly the following headings, depending on the runway in use:
  - (1) Runway 14: 290°
  - (2) Runway 32: Runway heading.

#### **2-1-3. IFR Departure Altitudes**

- a. IFR departure aircraft cleared via the DEVLN# SID must be instructed to maintain **9,000** feet and expect filed/amended altitude 10 minutes after departure.
- b. IFR departure aircraft not cleared via the DEVLN# SID must be instructed to maintain the following altitudes, dependent on the runway in use, and expect filed altitude 10 minutes after departure:
  - (1) Runway 14: 6,500 feet.
  - (2) Runway 32: 7,000 feet.

#### **2-1-4. VFR Departure Instructions**

- a. VFR departure aircraft must be handled in accordance with the ZLC Flight Strip Handling Directive (7210.3).
- b. Unless otherwise coordinated with LC, VFR departure aircraft requesting any direction of flight other than westbound must be instructed to expect one of the following VFR transitions/departures:
  - (1) Mountain Road Transition (Northbound or Southbound).
  - (2) I-15 Transition (Northbound or Southbound).
  - (3) MUFLAT Transition (Westbound)

## **Section 2. Ground Control**

### **2-2-1. Responsibilities**

- a. Monitor the GC frequency (121.600/275.80 MHz).
- b. Ensure all taxiing aircraft have the current ATIS.
- c. Mark FPSs in accordance with the ZLC Flight Strip Handling Directive (7210.3).
- d. Assume other ground control responsibilities, as assigned.
- e. Assume CD responsibilities when combined.

### **2-2-2. Area of Jurisdiction**

GC has jurisdiction of all taxiways in the movement area.

### **2-2-3. Traffic Flow**

GC must ensure all aircraft yield to aircraft exiting a runway, unless otherwise coordinated with LC.

### **2-2-4. ATIS**

GC must ensure all departing aircraft have the current ATIS information prior to communications transfer to LC.

### **2-2-5. Procedures**

- a. Hill AFB assigned aircraft and tow vehicles are authorized to taxi and tow on their respective ramps, including USFS on the Alert Ramp and other parking areas approved by AMOPS (i.e., hot pads) without prior ATC permission. ATC permission must be received before taxi or tow operations outside controlled ramp areas.
- b. Aircraft requiring de-arm shall establish and maintain contact with GC before taxiing from the de-arm area. The Tower establishes preferred taxi routing depending on ground traffic.
- c. Heavy aircraft are requested to taxi with outboard engines off or idle when possible to reduce FOD potential.

### **2-2-6. Departure**

- a. Control all taxiing aircraft except those on active runways.
- b. Approve pushback and start-up clearances.
- c. Ensure aircraft request taxi clearance with position and ATIS confirmation.

### **2-2-7. Arrivals**

- a. Aircraft not requiring de-arming after landing shall establish and maintain contact with GC before entering Taxiway Alpha.

## **Section 3. Local Control**

### **2-3-1. Responsibilities**

- a. Monitor the LC frequency (127.150/263.15 MHz).
- b. Provide control instructions to aircraft within the LC area of jurisdiction.
- c. Operate TDW/STARS equipment, as necessary.
- d. Marks FPSs in accordance with the ZLC Flight Strip Handling Directive (7210.3).
- e. Assume other local control or other cab responsibilities, as assigned.

### **2-3-2. Area of Jurisdiction**

LC has jurisdiction of the following:

- a. LC airspace, as depicted in Appendix 1.
- b. Runway 14/32.

### **2-3-3. Runway Selection**

Unless otherwise coordinated with Salt Lake City TRACON (S56), the active runway must be selected in accordance with the following:

- a. When the tailwind component is 10 knots or less, or the weather is below circling minima, Runway 14 should be used.
- b. When the tailwind component exceeds 10 knots from the north, Runway 32 should be used.

### **2-3-4. Use of Radar**

When using TDW/STARS, LC must:

- a. Comply with the provisions contained within FAA JO 7110.65 Chapter 5.
- b. Adjust the range of the radar to display a minimum of 15 NM.
- c. Display the HIF Tower List.

### **2-3-5. Departure Procedures**

- a. A "Release Request" (.RR) is required to be sent to the appropriate departure controller for all departing IFR aircraft.
- b. Departing aircraft must not climb above 6,300 feet until past the departure end of the runway when another aircraft is conducting the overhead maneuver.
- c. Departing VFR aircraft that will remain outside of the Ogden Class D airspace (see Appendix 1):
  - (1) Must be restricted at or below 7,500 feet until:
    - (a) Clear of the Hill Class D airspace (see Appendix 1), or;
    - (b) Communications transfer to S56.
  - (2) May be instructed to depart on course.
- d. Departing northbound VFR aircraft that will penetrate the Ogden Class D airspace must be restricted below 7,800 feet and communications transferred to Ogden LC (or other appropriate overlying controller).

### 2-3-6. Arrival Procedures

- a. IFR arrivals inbound on an IAP to Runway 14 utilize the Hill Arrival Corridor (see Appendix 1B). The Hill Arrival Corridor must not be utilized for any other operation, except as described in 5-8.
- b. VFR arrivals should enter the Hill Class D on a prescribed VFR transition, or over the Causeway (VPCWY) at 7,500 feet, to enter the traffic pattern.
- c. VFR arrivals or Category I/II IFR arrivals conducting a visual approach that execute a missed approach/go-around must be instructed to remain in the local traffic pattern, if capable of doing so.
- d. Except as described in subparagraph b., IFR arrivals that execute a missed approach/go-around must be instructed to comply with one of the following procedures/climbouts, depending on the arrival runway and organization status:
  - (1) Aircraft associated with the Virtual United States Air Force (vUSAF):
    - (a) Runway 14: Issue Layton climbout. The Layton climbout requires aircraft to "Track heading 139° until HIF 0.8 DME. Turn right heading 290, remain within HIF 5.7 DME until passing a heading greater than 256°. Cross HIF 0.8 DME at or below 6,300 feet. Climb and maintain 6,500 feet."
    - (b) Runway 32: Issue Riverdale climbout. The Riverdale climbout requires aircraft to "Track heading 319, cross HIF 1.6 DME at or below 6,300. Climb and maintain 7,000."

### PHRASEOLOGY-

*"TALON 14, execute Layton climbout."*

- (2) Aircraft not associated with the vUSAF:
  - (a) Runway 14: "Fly heading 290, climb and maintain 6,500."
  - (b) Runway 32: "Fly runway heading, climb and maintain 7,000."
- e. For aircraft that execute a missed approach/go-around that will not remain in the local traffic pattern, LC must:
  - (1) Send a Missed Notification (.MA) to the appropriate S56 sector that will be receiving the aircraft for re-sequencing.
  - (2) Ensure the aircraft does not climb above 6,300 feet until past the departure end of the runway when another aircraft is conducting the overhead maneuver.

### 2-3-7. Traffic Pattern Procedures

- a. Runway 14:
  - (1) Category I aircraft: Right closed traffic at 5,800 feet.
  - (2) Non-Fighter Category II/III aircraft: Right closed traffic at 6,300 feet.
  - (3) Fighter aircraft: Left or Right closed traffic (LC's discretion) at 6,800 feet.
- b. Runway 32:
  - (1) Category I aircraft: Left closed traffic at 5,800 feet.
  - (2) Non-Fighter Category II/III aircraft: Left closed traffic at 6,300 feet.
  - (3) Fighter aircraft: Left or Right closed traffic (LC's discretion) at 6,800 feet.

**NOTE-**

*Right breaks are standard for aircraft conducting the overhead maneuver on Runway 32.*

**2-3-8. Additional vUSAF Procedures**

To the maximum extent possible, LC must understand the following procedures for aircraft associated with the vUSAF:

- a. VFR traffic re-entering the traffic pattern following a missed approach/go-around must execute the following procedure depending on their arrival runway:
  - (1) Runway 14:
    - (a) Fly runway heading until the departure end of the runway, then make a climbing left turn (towards the mountains) to 6,800 feet and enter a left downwind.
  - (2) Runway 32:
    - (a) Fly runway heading to the HIF 2.0 DME, then make a right turn, crossing I-15 above 6,300 feet.
    - (b) The fighter will then turn north around the Freeport Center and continue so their base turn places them 1 NM south of the Ogden Airport (KOGD).
    - (c) Altitude and airspeed will normally be 6,800 feet and 300 knots, respectively.

**NOTE-**

*This procedure is depicted in Appendix 2A.*

- b. MUDFLAT recovery:
  - (1) Runway 14:
    - (a) Maintain 7,300 feet until past KOGD, then descend to pattern altitude (6,800 feet). Expect left turns.
  - (2) Runway 32:
    - (a) Proceed to the downwind leg (3 NM west of the runway) at 7,500 feet until abeam the runway threshold, then enter the overhead maneuver at 6,800 feet with right turns.

**2-3-8. VFR Procedures**

- a. VFR Missed Approach:
  - i. To RE-ENTER the Hill traffic pattern from a MISSED APPROACH, the aircraft will fly runway heading to the Hill TACAN 2 DME, then make a right turn crossing I-15 above 6,300 'MSL
  - ii. The fighter will then turn north around the Freeport Center and continue so his base turn places him 1 NM south of the Ogden Airport.
  - iii. Altitude and airspeed will be 6,800'MSL and 300 KIAS, respectively.
- b. VFR Departure Frequencies
  - i. Handoff to departure control as soon as clear of any possible conflicts
    - 1. S56 - 121.1 OR 135.5
  - ii. If it's a north VFR departure under 7,800 handoff to Ogden FCT

1. OGD TWR - 118.7
- c. VFR Entering Pattern
  - i. Enter Class D on prescribed VFR transition to enter traffic pattern to airport at 7,500 off of Causeway 4.
- d. VFR Flight Following
  - i. Ensure all aircraft requiring flight following have the appropriate scratch pad entry prior to handoff to departure.
- e. VFR Overhead Pattern
  - i. VMC to expedite recovery and landing of fighter-type aircraft The pattern consists of two 180-degree turns and resembles a race track
  - ii. It begins at approximately 3 to 5 NM from the approach end of the runway at an altitude of 6800' MSL
  - iii. Normal airspeed for flying initial is 300 KIAS with speed decreasing through the turns
  - iv. Over the approach end of the runway the pilot makes a 180 degree turn to downwind, then makes another 180 degree turn to complete the base to final portion of the pattern
  - v. The pattern is concluded either by a full stop or missed approach

#### **2-3-9. VFR Traffic Pattern Runway 32:**

- a. Light Civil A/C Use Left Closed Pattern @ 5,800
- b. Larger Than Fighter Type Aircraft Will Use Left Closed Pattern @ 6,300
- c. For Fighter Type Aircraft:
  - i. Traffic Pattern Alt: 6,800
  - ii. Left-Hand Or Right-Hand Circuits On All Runways (Tower's Choice)
- d. Right brakes are standard for Runway 32.
- e. The aircraft executing the MUDFLAT recovery will initially proceed to the downwind leg 3 miles west of the runway at 7,500 until it abeam the landing threshold, then enter the overhead pattern at 6,800 right turns.
- f. Fly over Freeport center on the outside downwind
  - i. Left reentries: Aircraft will enter an outside downwind 3NM west of the field at 6,800 and report a 3-mile initial.

#### **2-3-10. VFR Traffic Pattern Runway 14:**

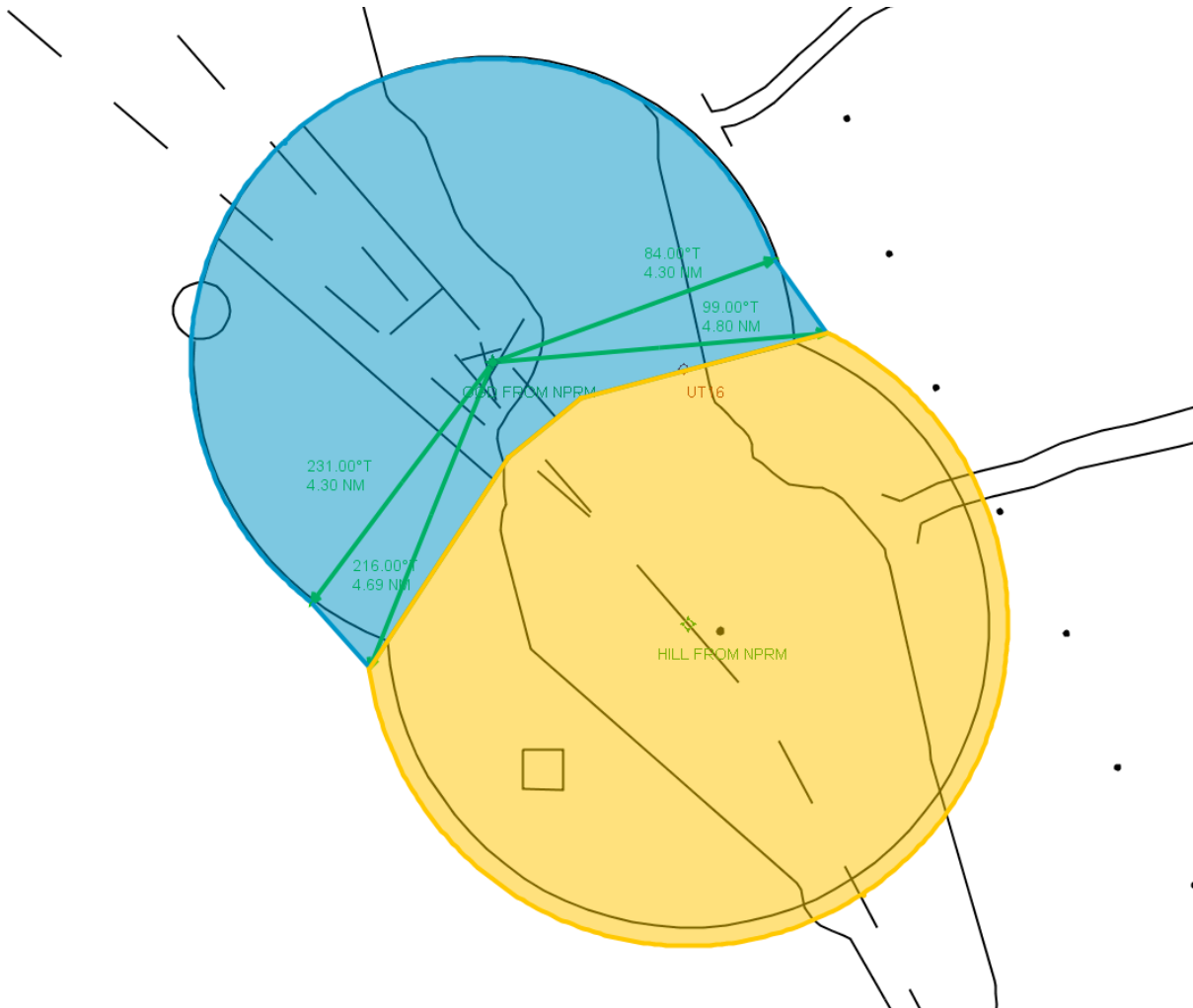
- a. Light Civil A/C Use Right Closed Pattern @ 5,800
- b. Larger Than Fighter Type Aircraft Will Use Right Closed Pattern @ 6,300
- c. For Fighter Type Aircraft:
  - i. Traffic Pattern Alt: 6,800
  - ii. Left-Hand Or Right-Hand Circuits On All Runways (Tower's Choice).
- d. The aircraft executing the mudflat recovery will initially maintain 7,300 until past Ogden, then descend to pattern altitude 6,800 left turns.
- e. To remain in a CLOSED PATTERN, after a low/missed approach, the pilot will continue to the end of the runway and make a climbing left turn (towards the mountains) to an altitude of 6800' MSL.

- i. This will place him in the pattern on a left downwind.
- ii. Aircraft departing Weber Canyon should be aware of aircraft in this traffic pattern due to the aircraft being belly up to the mountains



## Appendix 1. Airspace

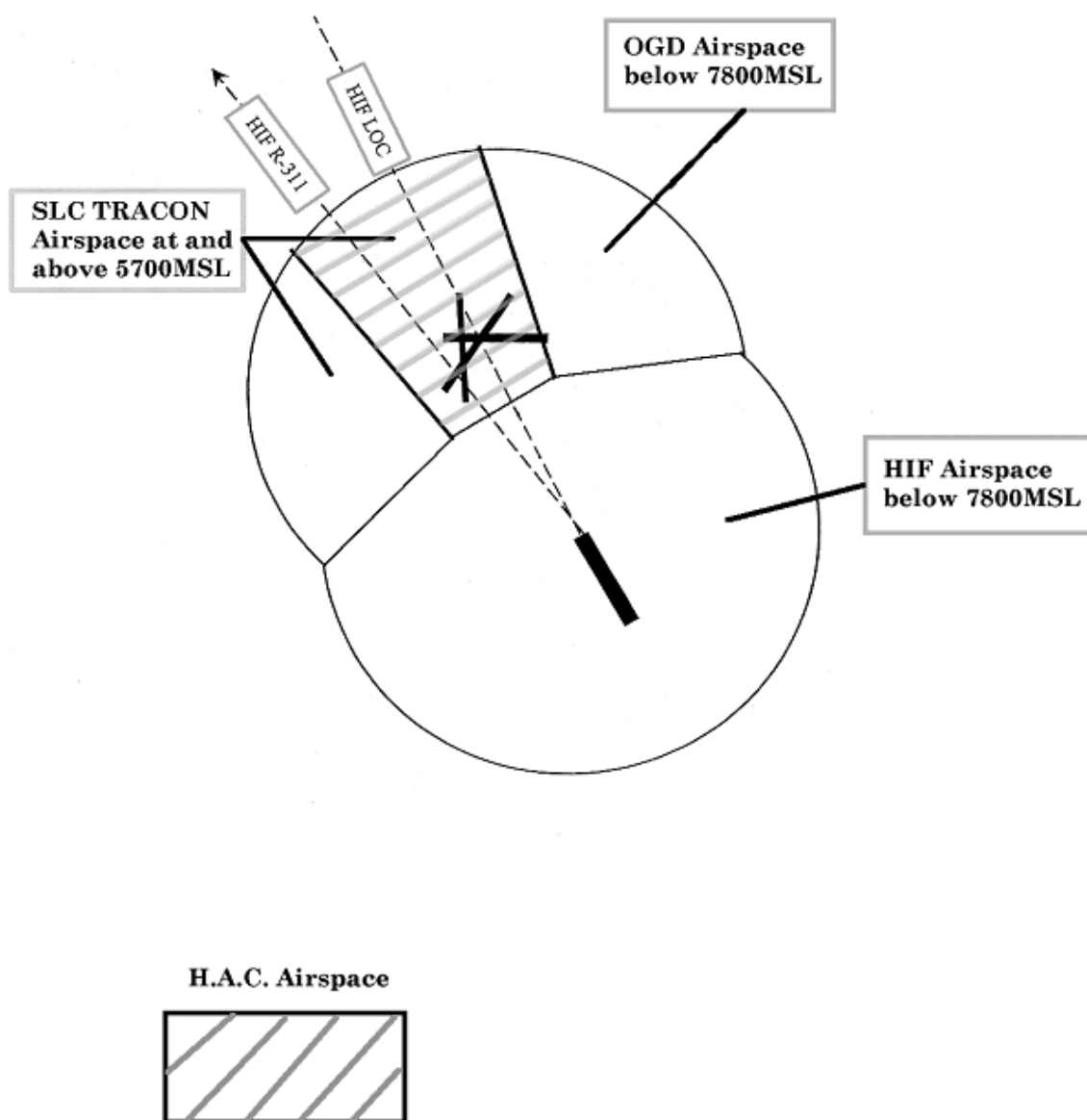
### Appendix 1A. LC Airspace



#### **NOTE-**

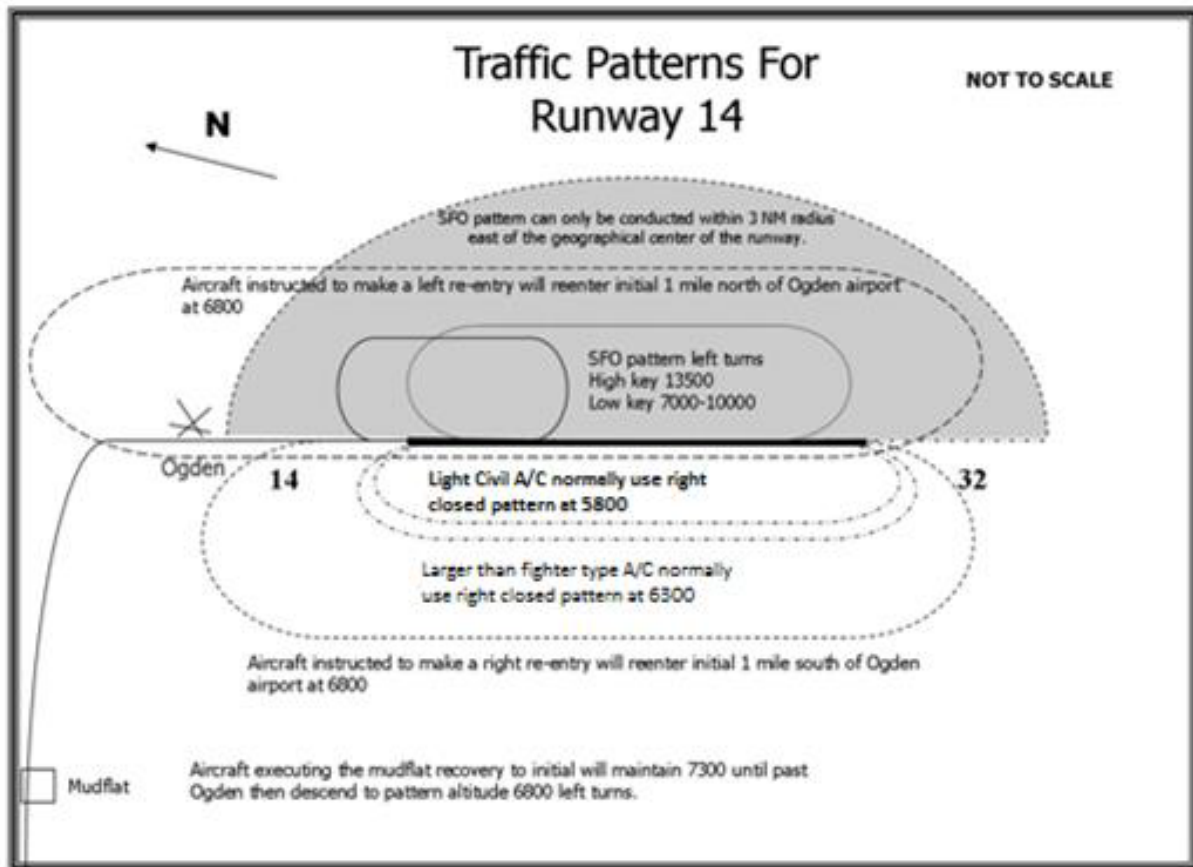
The yellow-colored region represents LC airspace below 7,800 feet MSL. The blue-colored region is delegated to Ogden FCT below 7,800 feet MSL.

## Appendix 1B. Hill Arrival Corridor (HAC)

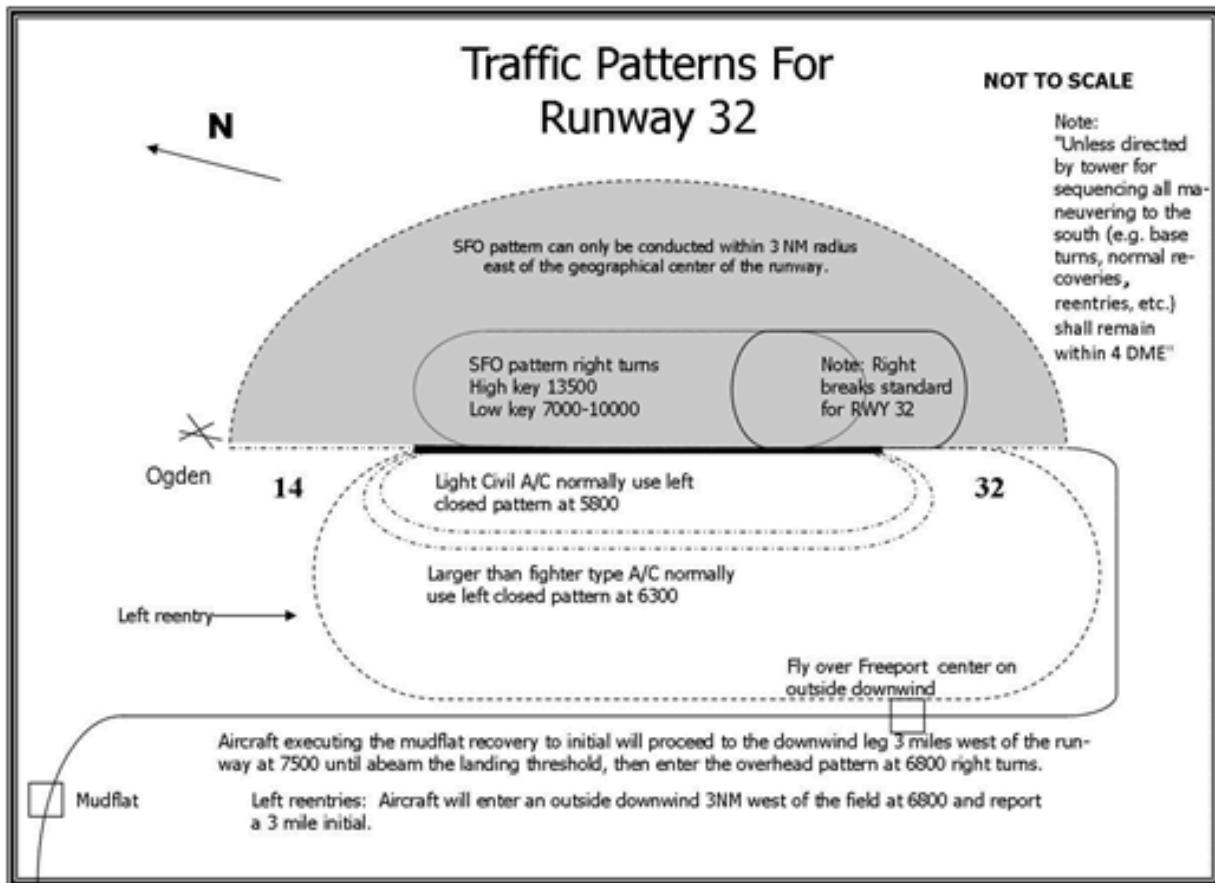


## Appendix 2. Traffic Patterns

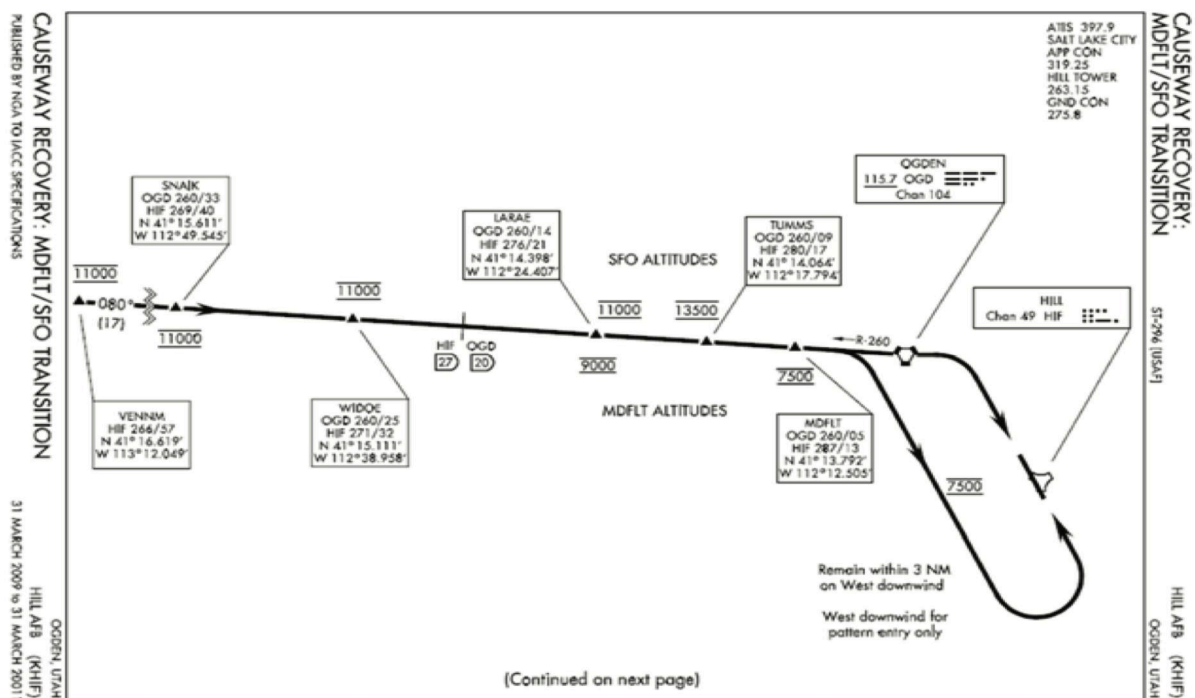
### Appendix 2A. Runway 14 (vUSAF Only)



## Appendix 2B. Runway 32 (vUSAF Only)



### Appendix 3A. Mudflat Recovery (Chart)



### Appendix 3B. Mudflat Recovery (VFR Map)

